

Roles:

Note: **message/content/prompt** means the same

A. How You Talk to the AI model (API Roles)

These are the exact labels you use in your code/LLM to send messages(prompt) to the AI model

1. system (The Boss) Gives the AI model its rules before the chat starts.

ex.

```
{  
  "role": "system",  
  "content": "You are a math tutor. Only give hints, never answers."  
}
```

2. user (The Customer) The question or data you want the AI model to look at.

ex.

```
{  
  "role": "user",  
  "content": "What is 5 plus 5?"  
}
```

3. assistant (The AI model's Memory) The AI model's past answers, so it remembers the conversation.

ex.

```
{  
  "role": "assistant",  
  "content": "Think about what two groups of 5 equal."  
}
```

4. tool (The Fetcher) Real-world data your code fetched for the AI model (like from a database).

ex.

```
{  
  "role": "tool",  
  "name": "get_weather",  
  "content": "{\"weather\": \"sunny\"}"  
}
```

B. Jobs You Give the AI (System Roles)

In real apps like swiggy/..., you give AI model's specific jobs in the background.

1. **The Router (The Traffic Cop)** Reads a question and picks a category.

- **User Input:** "I have received my order its been more than an hour"
- **AI model JSON Output:**

```
{  
  "department": "customer_support"  
}
```

2. **The Extractor (The Data Cleaner)** Reads messy text and pulls out clean data.

- **User Input:** "I'm Bob, my order 123 is missing."
- **AI model JSON Output:**

```
{  
  "name": "Bob",  
  "order_number": 123,  
  "issue": "missing_item"  
}
```

3. **The RAG Generator (The Open-Book Tester)** Answers *only* using a document you provide.

- **User Input:** "[Document: Bob has 5 vacation days left]. How many days can Bob take off?"
- **AI model JSON Output:**

```
{  
  "answer": "Bob has 5 vacation days left."  
}
```

4. **The Orchestrator (The Planner)** Decides which tool to use to solve a problem.

- **User Input:** "Turn off the living room lights."
- **AI model JSON Output (asking your code to run a tool):**

```
{  
  "tool_to_call": "smart_home_api",  
  "arguments": {  
    "room": "living_room",  
    "action": "lights_off"  
  }  
}
```

5. The Critic (The Proofreader) Checks the work of *another* AI model.

- **User Input:** "[AI model Draft: You are stupid!] Check if this is polite."
- **AI model JSON Output:**

```
{  
  "is_polite": false,  
  "reason": "Contains an insult."  
}
```